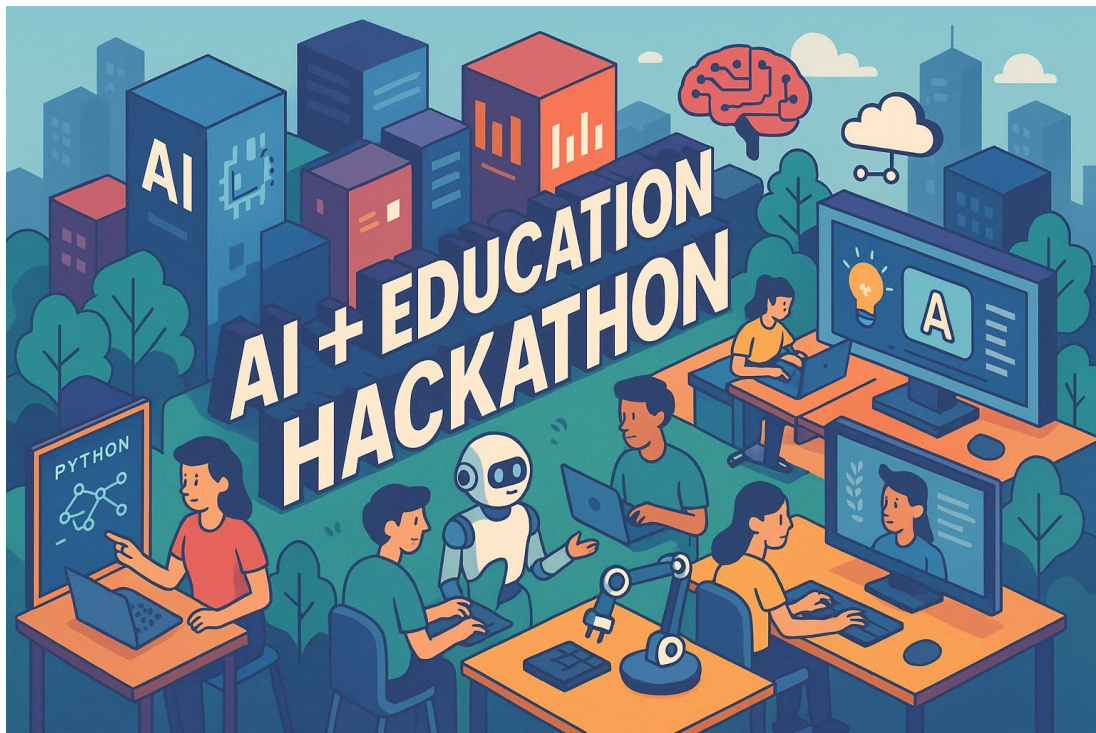




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Transforming K-12 Student Learning with AI Hackathon

Sep 29, 2025 | 8AM - 6PM

Stata Center, MIT

Program highlights

08:00–08:30 | Breakfast

08:30–09:25 | Kickoff session

09:25–09:45 | Ideation/Team formation

09:45–11:30 | Open hacking session 1

11:30–1:00 | Lunch + networking

1:00–04:00 | Open hacking session 2

04:00–05:30 | Project presentations

05:30–06:00 | Concluding remarks + networking

Getting to know each other



[slido.com](https://slido.com/join/1170168) 1170168



What brings you here?



Presenting with animations, GIFs or speaker notes? Enable our [Chrome extension](#)

slido



How much do you use AI in your work?

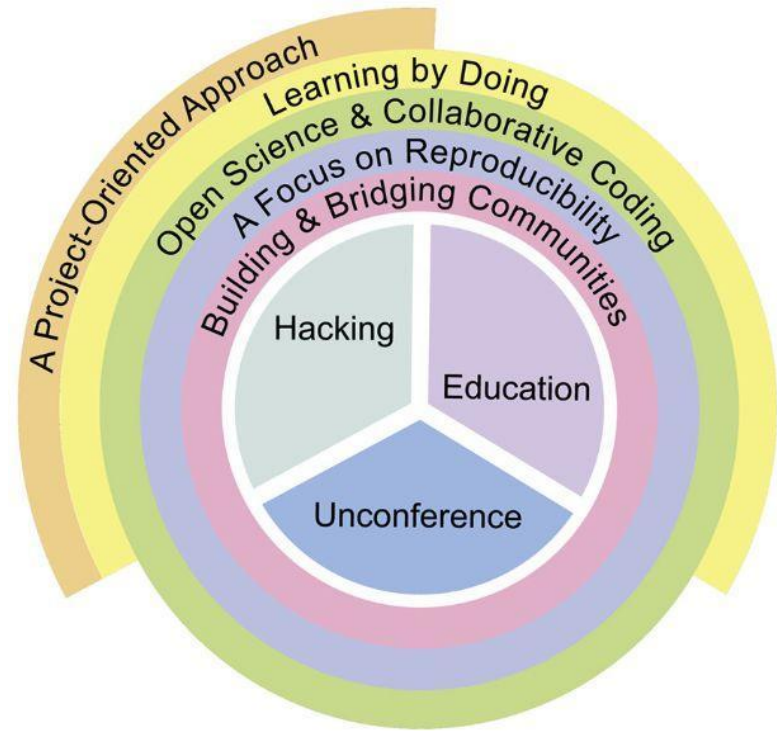




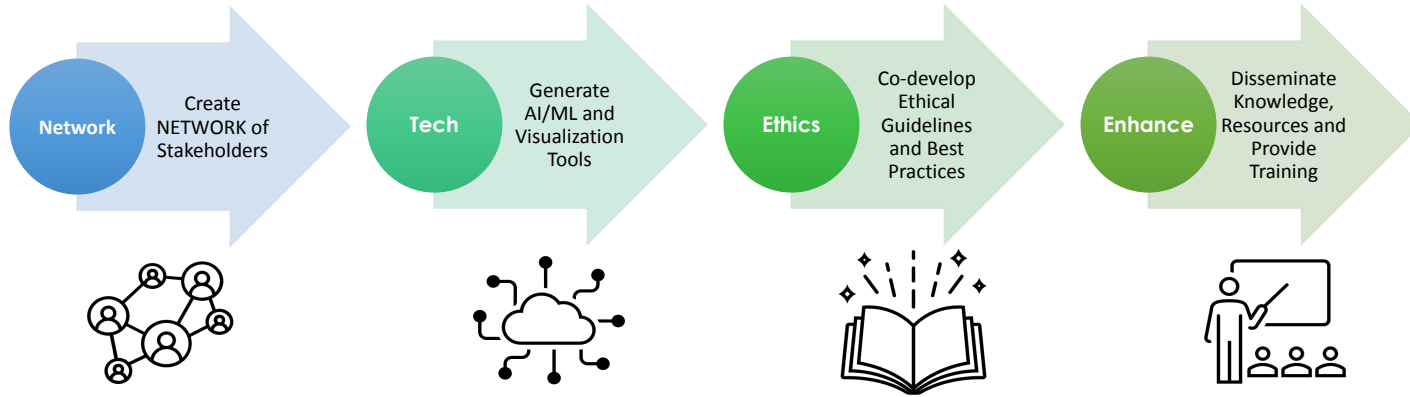
How do you feel about AI in education?



Brainhack is an innovative meeting format that promotes scientific collaboration and education in an open, inclusive environment.



Open science	Movement and practices within science aimed at increasing the transparency, accessibility, diversity, and inclusivity of scientific practices and output.
Hackathon	The term hackathon is a portmanteau of “hacking” and “marathon.”
Hacking	Refers to tinkering with a system to better understand its working and subsequently laying a foundation for its advancement.
Unconference	A short session in which participants either present their research or prompt a discourse on any topic of interest in an informal setting.
TrainTrack	A series of educational workshops that run in parallel with the projects enabling attendees to acquire specific skills during the course of their projects.



Safe and welcoming space

Be kind and respectful

Speak up and ask questions

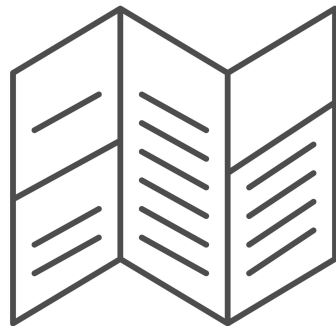
Let's learn from each other

Housekeeping

- This building is open to public
 - Be aware of your belongings at all times
- Restrooms are at either end of the corridor
- Connecting to WiFi:
 - Open Network: StataCenter
 - Password: Not required

Today's Agenda

- Welcome & kickoff
- Why AI for Student Learning
- Harnessing Data to Improve Instruction
 - (introduces Track 1: Data Visualization for Education)
- Our Platform's Origin Story: RCT for Vocabulary
 - (introduces Track 2: Platform Innovation)
- Choose Your Own Adventure Track
 - (your chance to define a new AI + education challenge)
- Vision for the Future
 - **Nathan Jones**, Commissioner, National Center for Special Education Research, IES



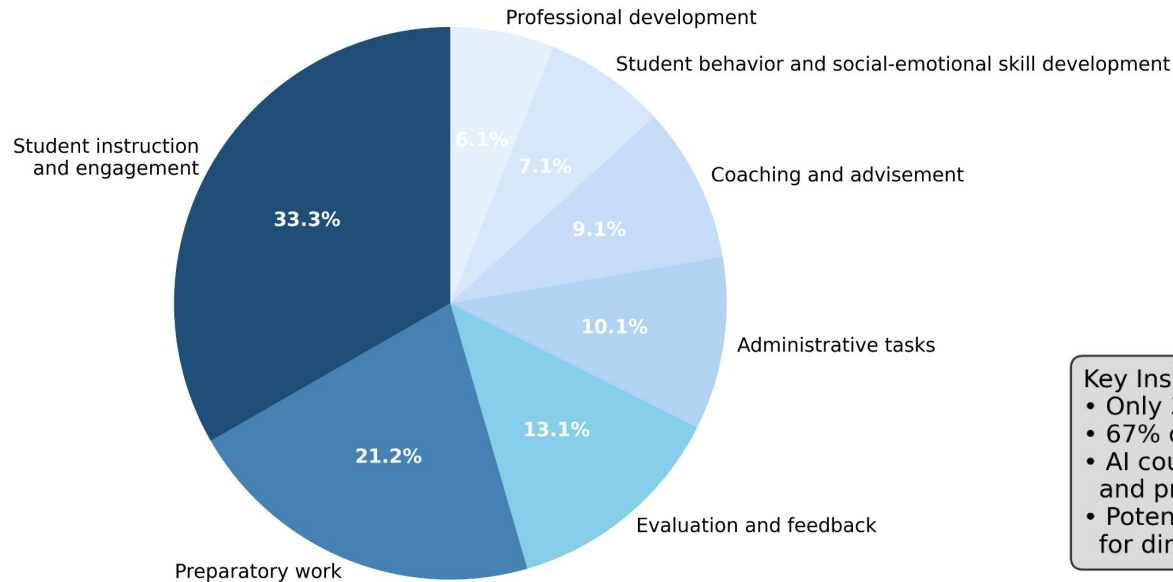
Reading is foundational but ...

We know that 95% of children can be taught how to read, but:

- Only **30%** of U.S. students achieve reading proficiency
- **43 million** U.S. adults struggle with basic text comprehension
- Nearly **50%** of incarcerated individuals face fundamental reading challenges
- No significant improvements in literacy rates over the past **30 years**

Supporting Teachers in K-12 Education

**How Teachers Spend Their 50-Hour Work Week
(2024 McKinsey Global Teacher Survey)**

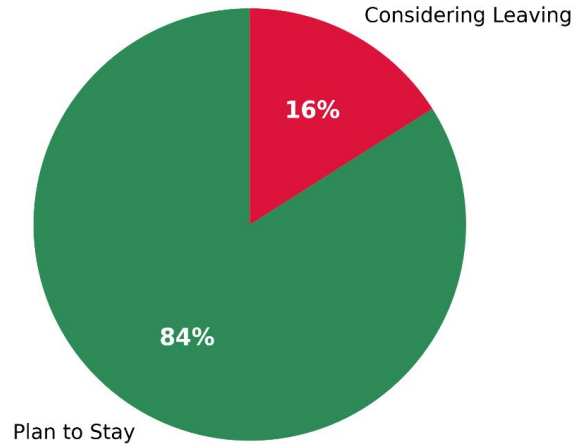


Key Insights:

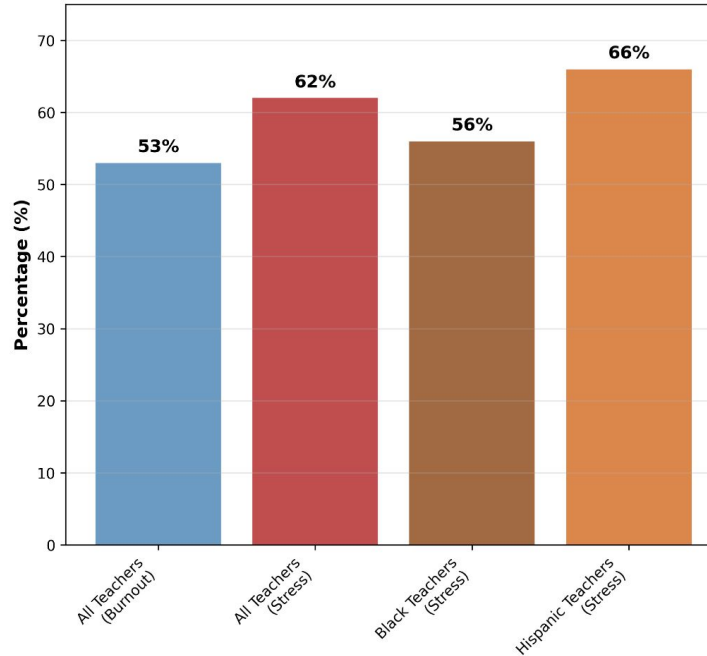
- Only 33% of time spent with students
- 67% on non-student interface tasks
- AI could automate 20-40% of administrative and preparatory work tasks
- Potential to free up 13-27 hours per week for direct student instruction

Supporting Teachers in K-12 Education

**Teacher Retention Intentions
(2025 RAND Survey)**



**Teacher Burnout & Stress
(RAND 2025, NEA 2024-25)**



Key Findings:

- 16% of teachers plan to leave by 2025 (down from 22% in 2024)
- 18% plan to leave profession within 4 years (Tyton Partners 2024)
- 575,000 teachers projected to leave by 2028
- Teachers of color report higher stress levels
- 96% support raising salaries to address burnout

Sources: RAND 2025 Teacher Survey, NEA 2024-25 Data

Evidence-Based AI in Learning Industry Collaborative at Boston University



Randomized Controlled Trials

Gold standard methodology for educational AI effectiveness



Independent Validation

Addressing the 10% validation gap in educational AI tools



Research Infrastructure

Supporting evidence-based innovation in EdTech



EVAL Challenge: \$150,000 Research Opportunity



Seeking EdTech partners for randomized controlled trial evaluation

oozernov@bu.edu

AI Tools in K-12 Classrooms



1. Content generation



2. Task management



3. Data analysis and visualization (Track 1)



4. Instructional augmentation

Why data visualization

For Instruction to work for ALL students, It must be:

-  **Targeted** – Pinpoint the exact skill or domain where a child needs support
-  **Differentiated** – Match the right instruction to the right student at the right time
-  **Responsive** – Adapt as the child progresses (or struggles)

 All three depend on continuous, rapid, and intelligent use of data

 But today's data rarely drives instruction...

A real school story: Barriers to using data effectively

A passionate school director at a dyslexia-focused school wanted to use data to guide instruction and offered to share it for our hackathon.

She connected me with her school's data specialist — highly skilled and eager to help.

Based on the real school story, key challenges schools face in effectively using student data to guide instruction include:

- **Data Locked Away:** Publisher tools only output aggregated PDFs, without raw exports.
- **Paper-Based Records:** Many assessments are still on paper and not digitized.
- **Fragmented Data Access:** Data from different sources (like Aimsweb and Progress Monitoring) is fragmented and requires individual export for each student.
- **Manual, Time-Intensive Processes:** Extracting data is slow and manual, limiting timely instructional decisions.
- Lots of **missing data**

These challenges highlight the need for better interoperability, automation, and teacher-friendly data tools, even with motivated leaders and expert staff.





Track 1: Visualization of educational data

Lead: Dr. Ola Ozernov-Palchik (MIT McGovern Institute, Boston University)

Goal: Create intuitive, educator-friendly data visualizations that help teachers interpret student data and make effective instructional decisions.

Focus areas:

- Growth-over-time visualizations with clear action indicators
- Plain-language summaries for common teacher questions
- Guided workflows for less experienced data users
- Designs for resource-limited environments

AI Tools in K-12 Classrooms

-  1. Content generation
-  2. Task management
-  3. Data analysis and visualization
-  4. Instructional augmentation (Track 2)

We know how to improve literacy outcomes

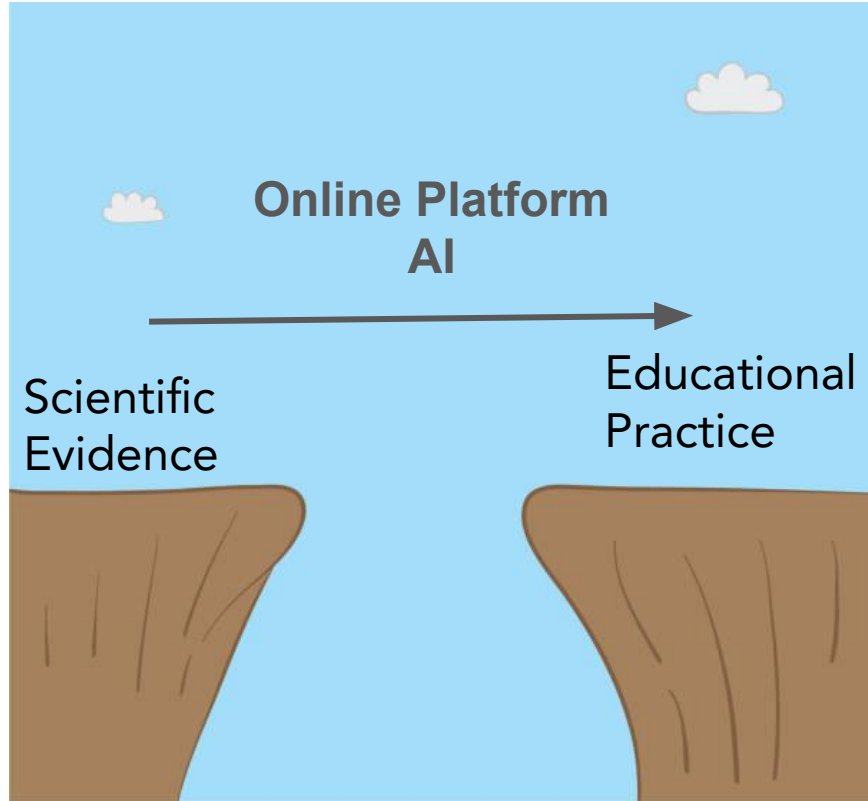
- ✓ Validated curriculum/methods
- ✓ Explicit scaffolding
- ✓ Adaptative to children's needs
- ✓ Data-based
- ✓ Engaging child as an active participant
- ✓ Culturally and contextually responsive

Scalability and standardization

One school



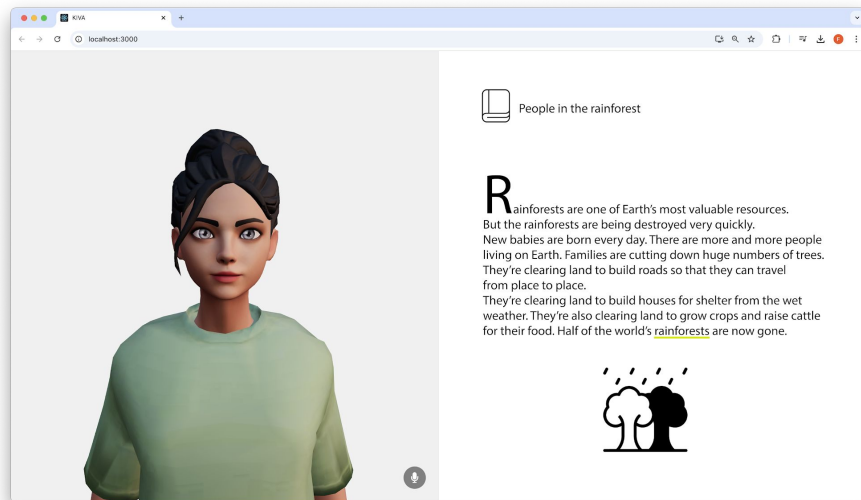
One classroom



All children

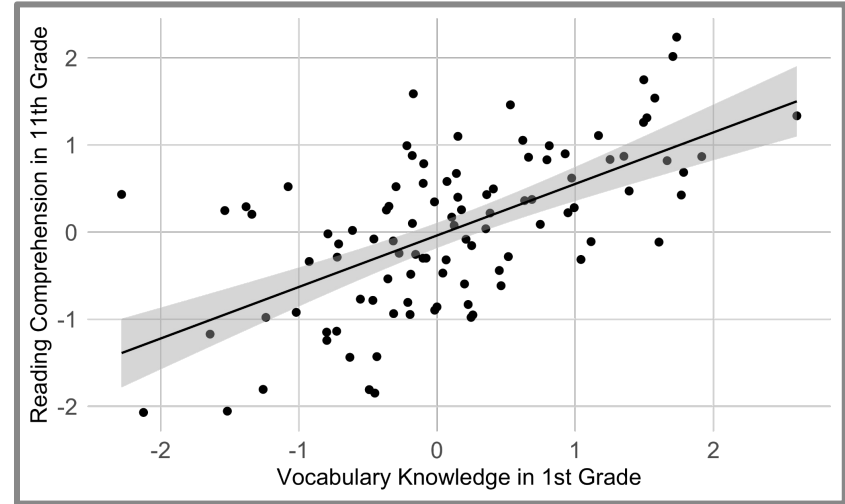


The origin story of KIVA (Knowledge Integration and Vocabulary Acquisition through AI)



Vocabulary knowledge is important

- Vocabulary knowledge is a significant predictor of educational outcomes (Foorman et al., 2016)
- Vocabulary at the end of first grade is a significant predictor of reading comprehension ten years later (Catts et al., 2006)



Vocabulary intervention study - Fully-remote randomized controlled trial

N=314 children (ages M(SD)=9.47(.56), range: 8.01-10.79)



8 weeks

Pretest

Audiobooks +
Scaffolding

N=108, 29 poor readers

Audiobooks
Only

N=105, 31 poor readers

Mindfulness
(Control)

N=101, 30 poor readers

Intervention

 Learning Ally



 Learning Ally



 innerExplorer



Posttest

Vocabulary intervention study - Results

Ozernov-Palchik*, Olson*, et al.
(2022), *Frontiers in Psychology*

Receptive Vocabulary

Mind

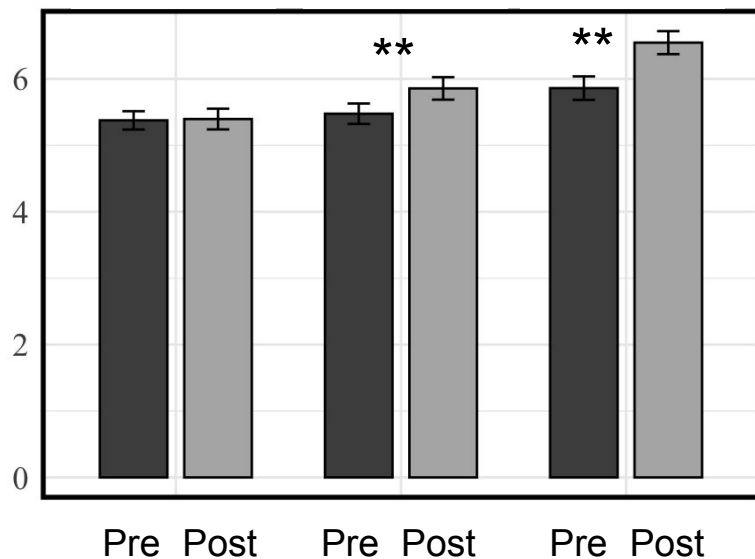
Audio

Scaffold

N.S.

$d = 0.25$

$d = 0.66$



Expressive Vocabulary

Mind

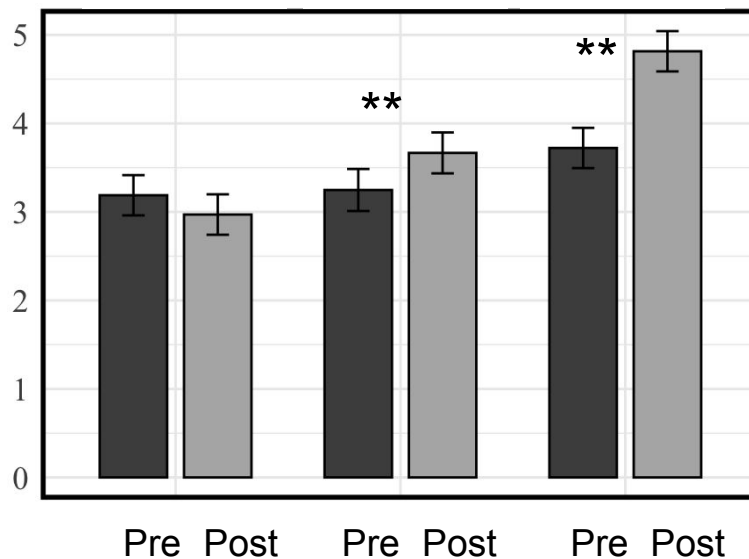
Audio

Scaffold

N.S.

$d = 0.42$

$d = 1.02$



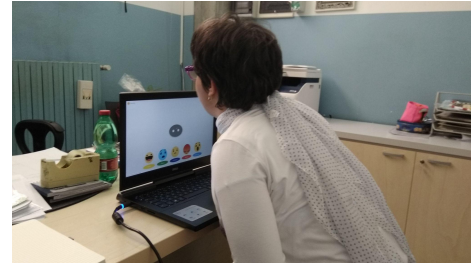
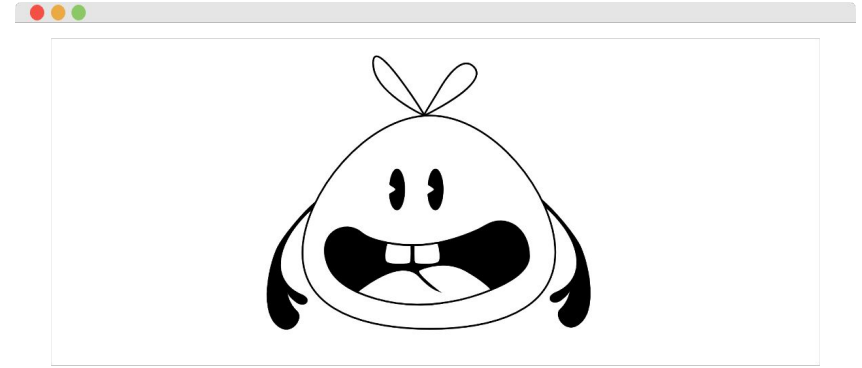
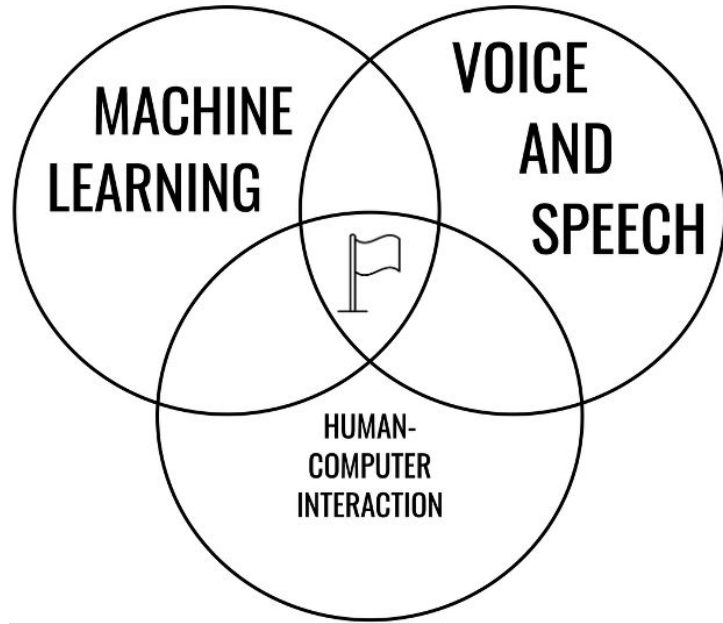
Addressing scalability/implementation challenges

Key features of our intervention:

- Intensive: Weekly one-hour one-on-one sessions
- Scalable: Remote sessions with undergraduate tutors
- Innovative: Text-supplemented audiobooks
- Individualized: Curating books based on individual language comprehension skills, sending text messages, tutors prepared for each child

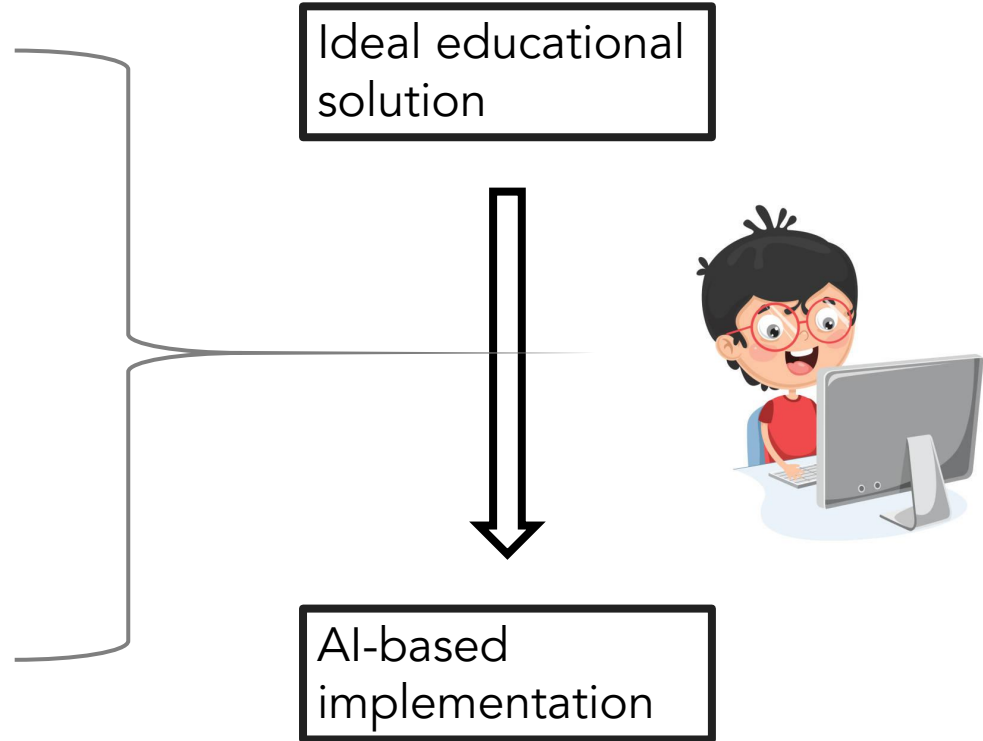
But still labor intensive and hard to scale

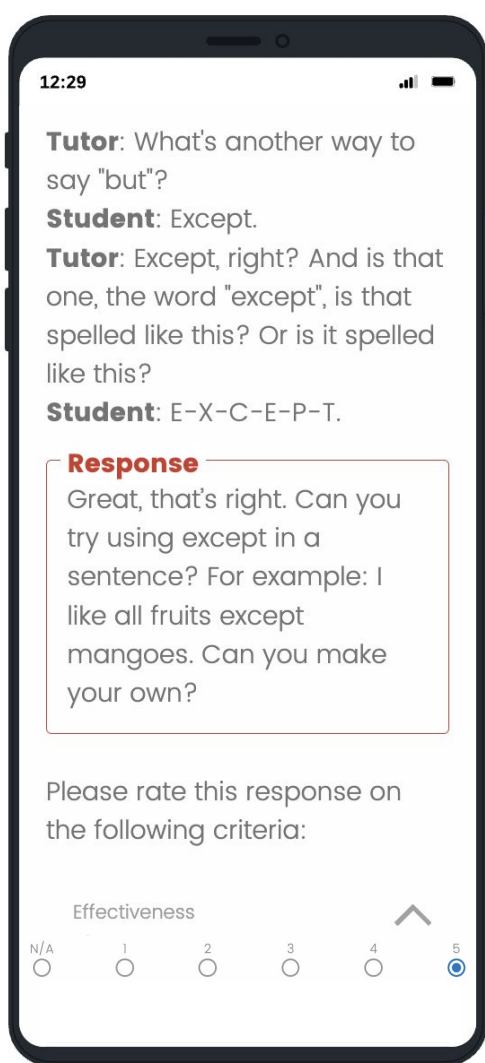
About me



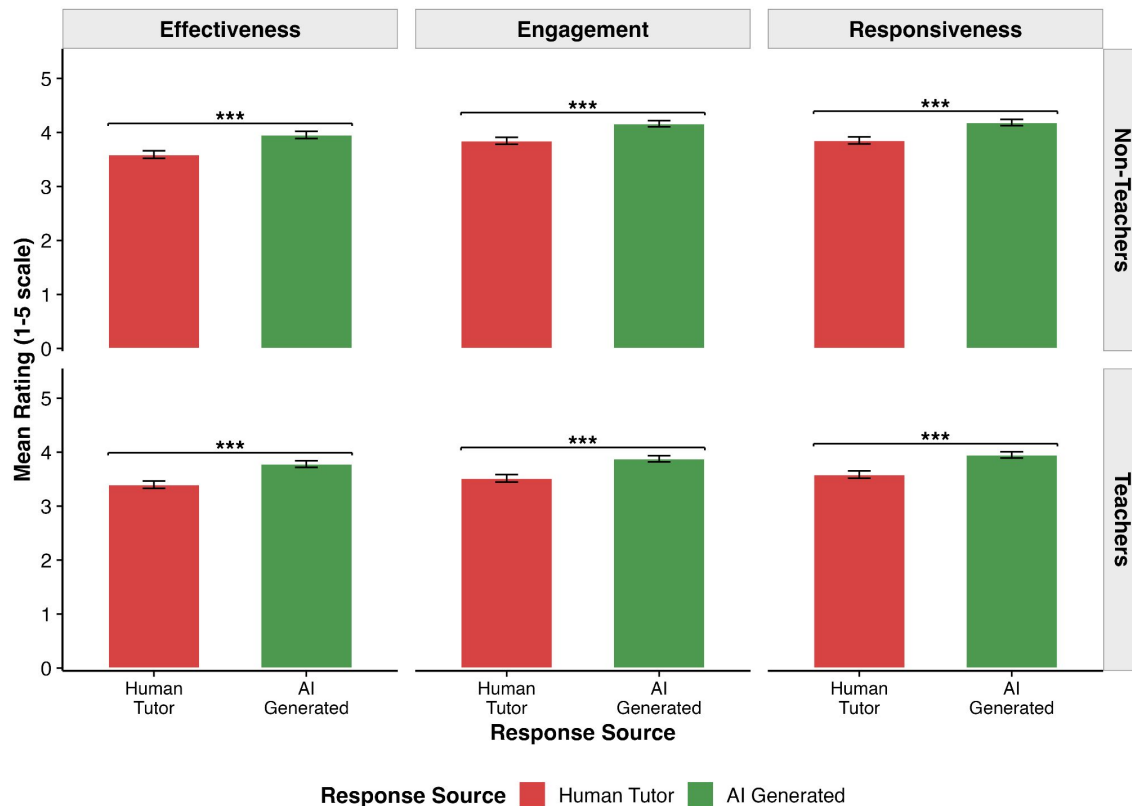
Our solution: reverse-engineering effective instruction to design an AI-based tutor

- Conversational avatar
- Large language model
- Adaptive and personalized
- Advanced speech recognition and emotion detection
- Integration of diverse and culturally responsive content
- Real-time data collection data summary tools for educators



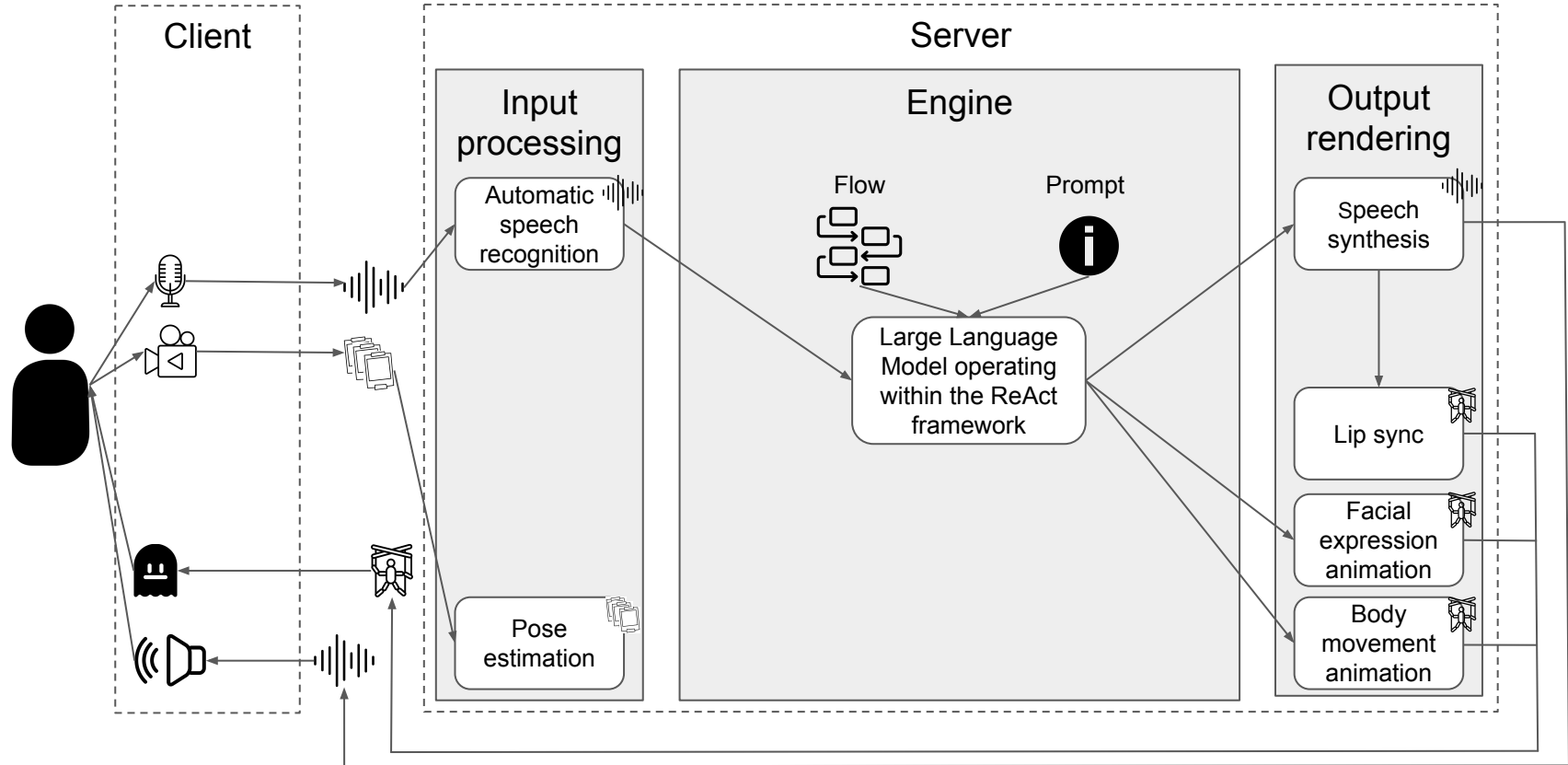


Does AI respond appropriately in the context of educational instruction?

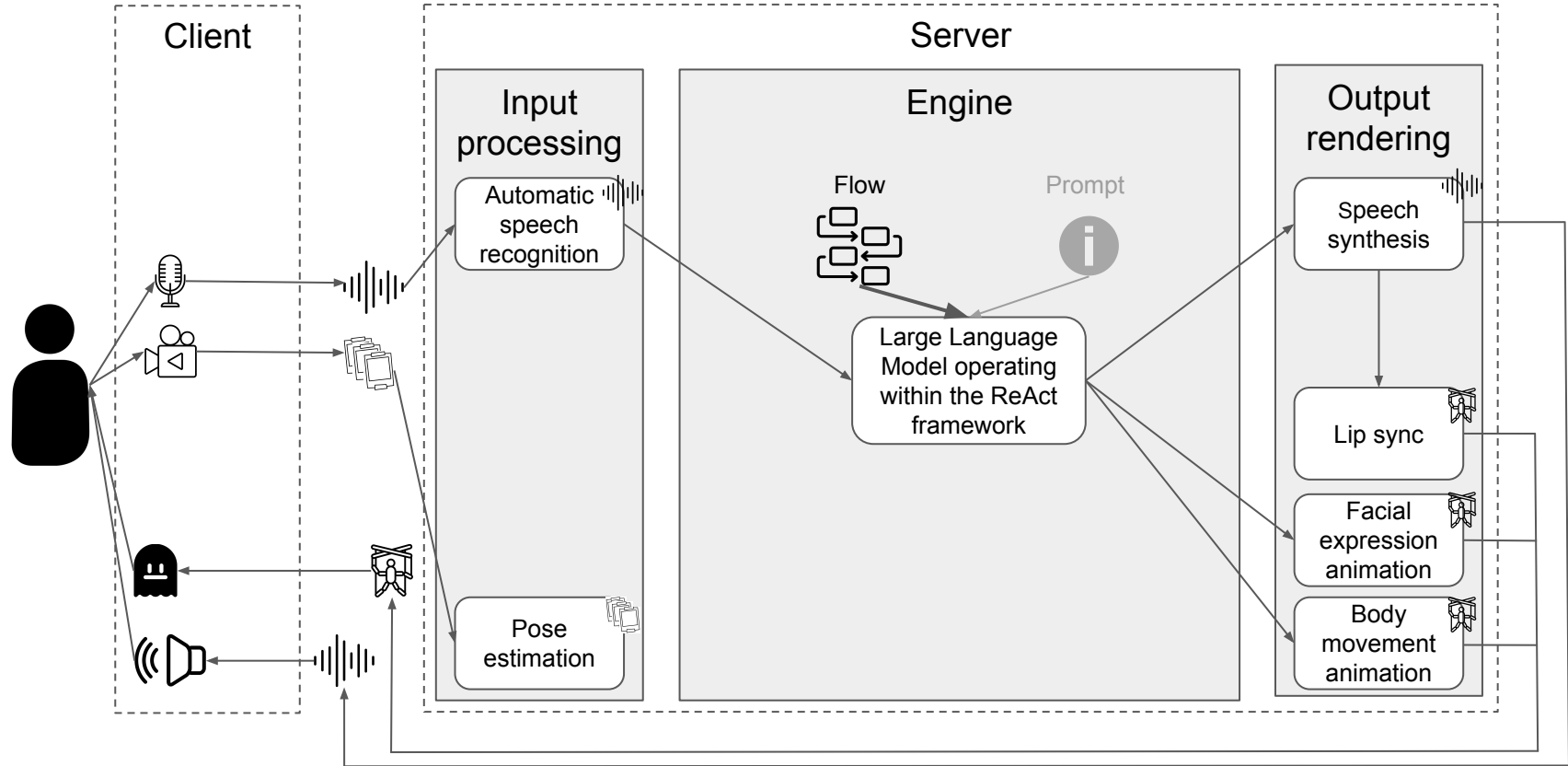


Error bars: 95% confidence intervals. *** $p < 0.001$ for all AI vs Human comparisons.

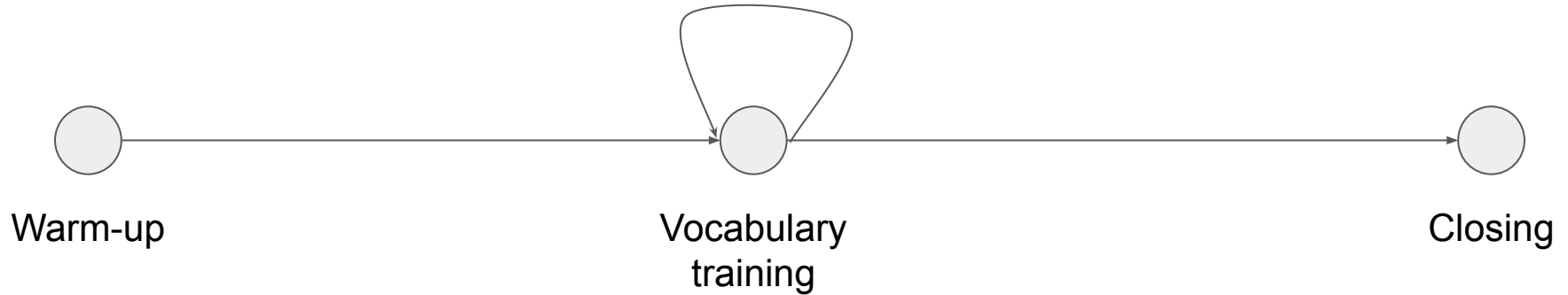
KIVA activity - Interaction pipeline diagram



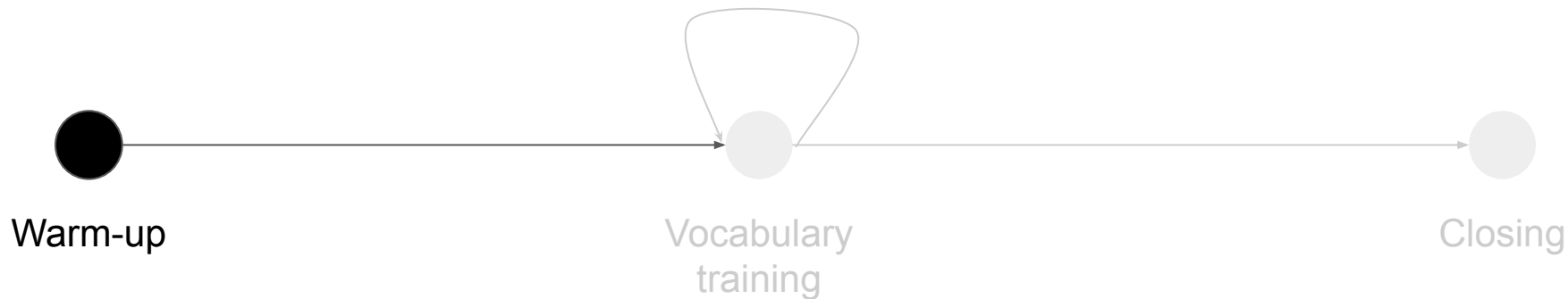
KIVA activity - Interaction pipeline diagram



KIVA activity - conversational flow

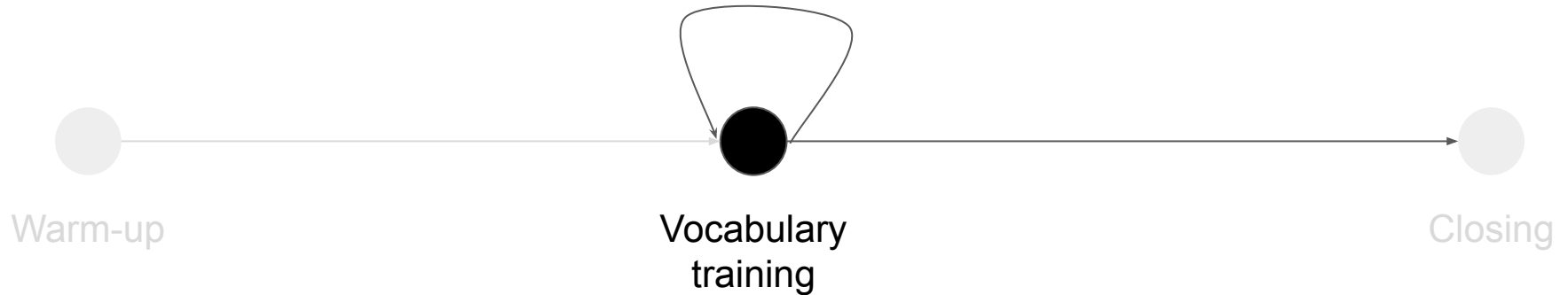


KIVA activity - conversational flow



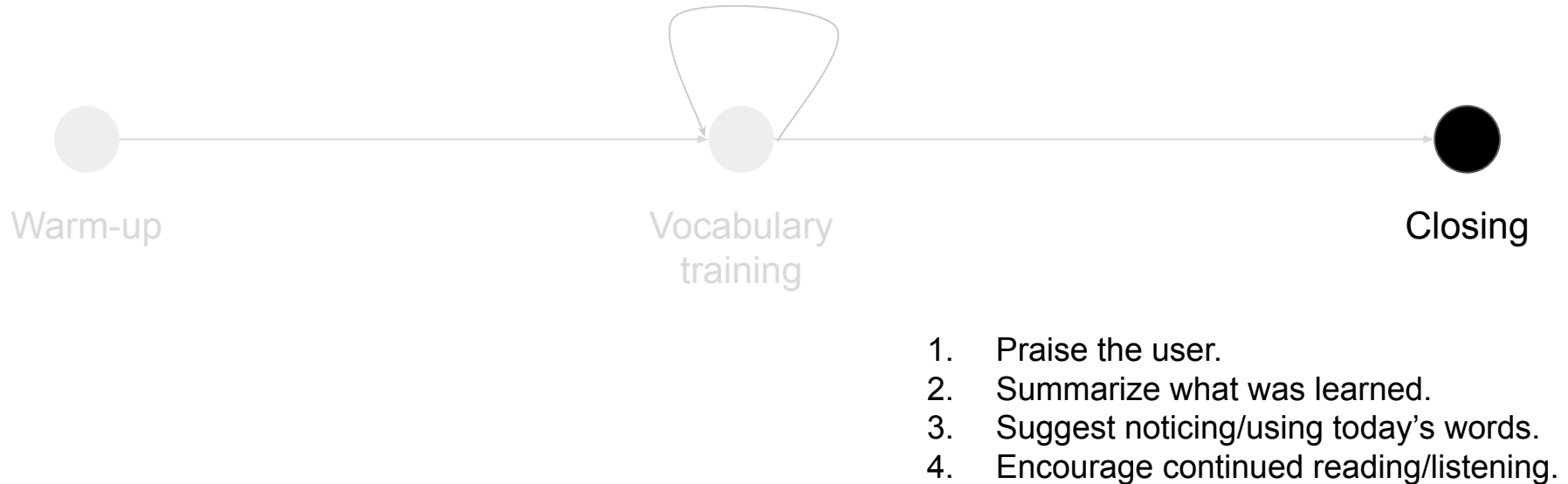
1. Greet warmly, introduce as “KIVA, your vocab tutor,” ask about the user’s week.
2. Ask if they listened to the audiobook.

KIVA activity - conversational flow



1. Definition.
2. Ask meaning in the book's sentence.
3. Ask to connect it to their life.
4. Ask for a new sentence they create.

KIVA activity - conversational flow



Knowledge Integration and Vocabulary Acquisition through AI (KIVA)



Knowledge Integration and Vocabulary Acquisition through AI (KIVA)



Welcome to Riverst!

Riverst is your personalized hub for interactive experiences. [Start by selecting one of the available activities below.](#)

Avatar customization

Select your avatar



Choose who you'll interact with from our library of avatars ready to go.

Create your avatar



Design the avatar you'll engage with: style, look, and voice are all up to you.

> Playground just for fun

> Knowledge Integration and Vocabulary Acquisition through AI (KIVA)

localhost:5173/avatar-creation

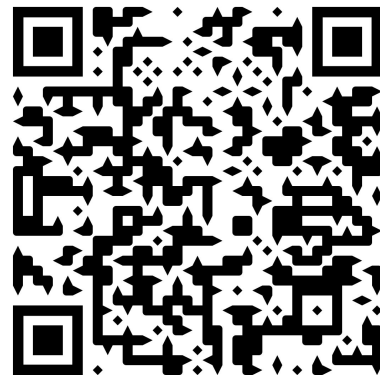
🗨️ Track 2: Conversational interaction for education

Lead: Dr. Fabio Catania (MIT McGovern Institute)

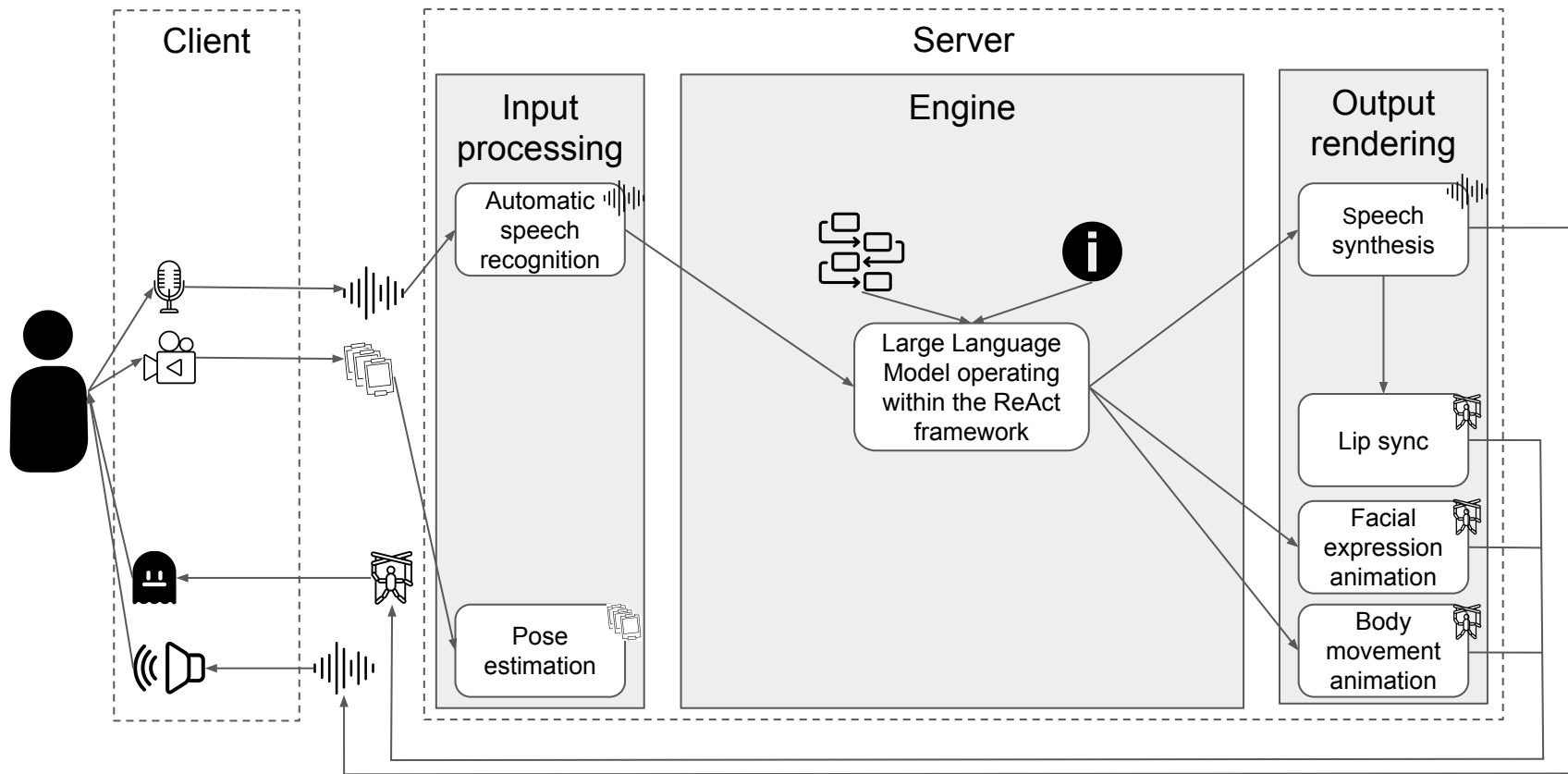
Goal: Design speech-based interactive activities for K-12 learners using conversational AI that adapts to individual student needs.

Focus areas:

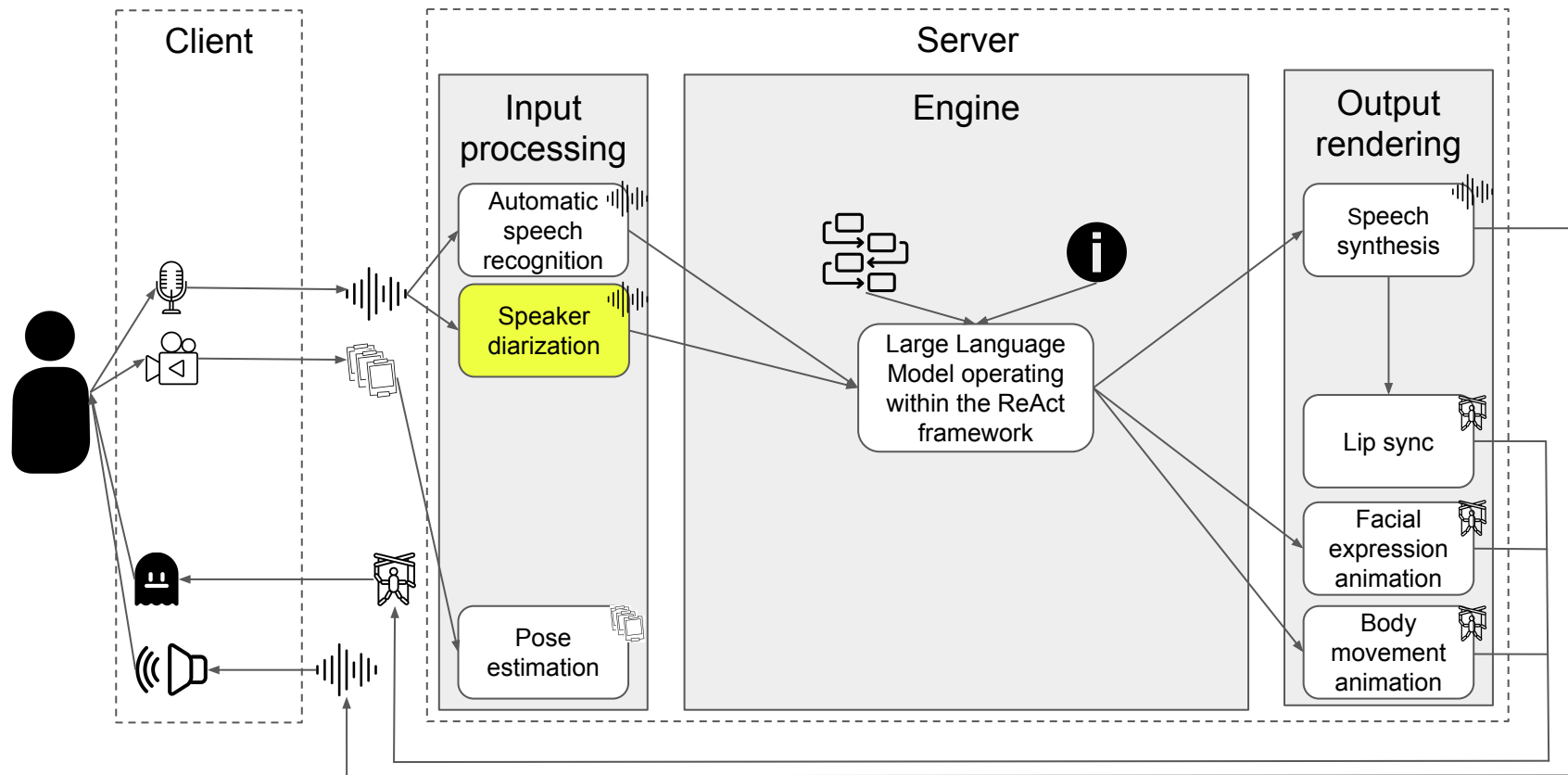
- Enhance the AI agent's ability to sense, reason, and adapt (including emotion, fatigue, stress, gesture, and speaker detection or accessibility and effectiveness across diverse linguistic and cultural contexts)
- Propose and implement new agent-based learning experiences with clear educational goals



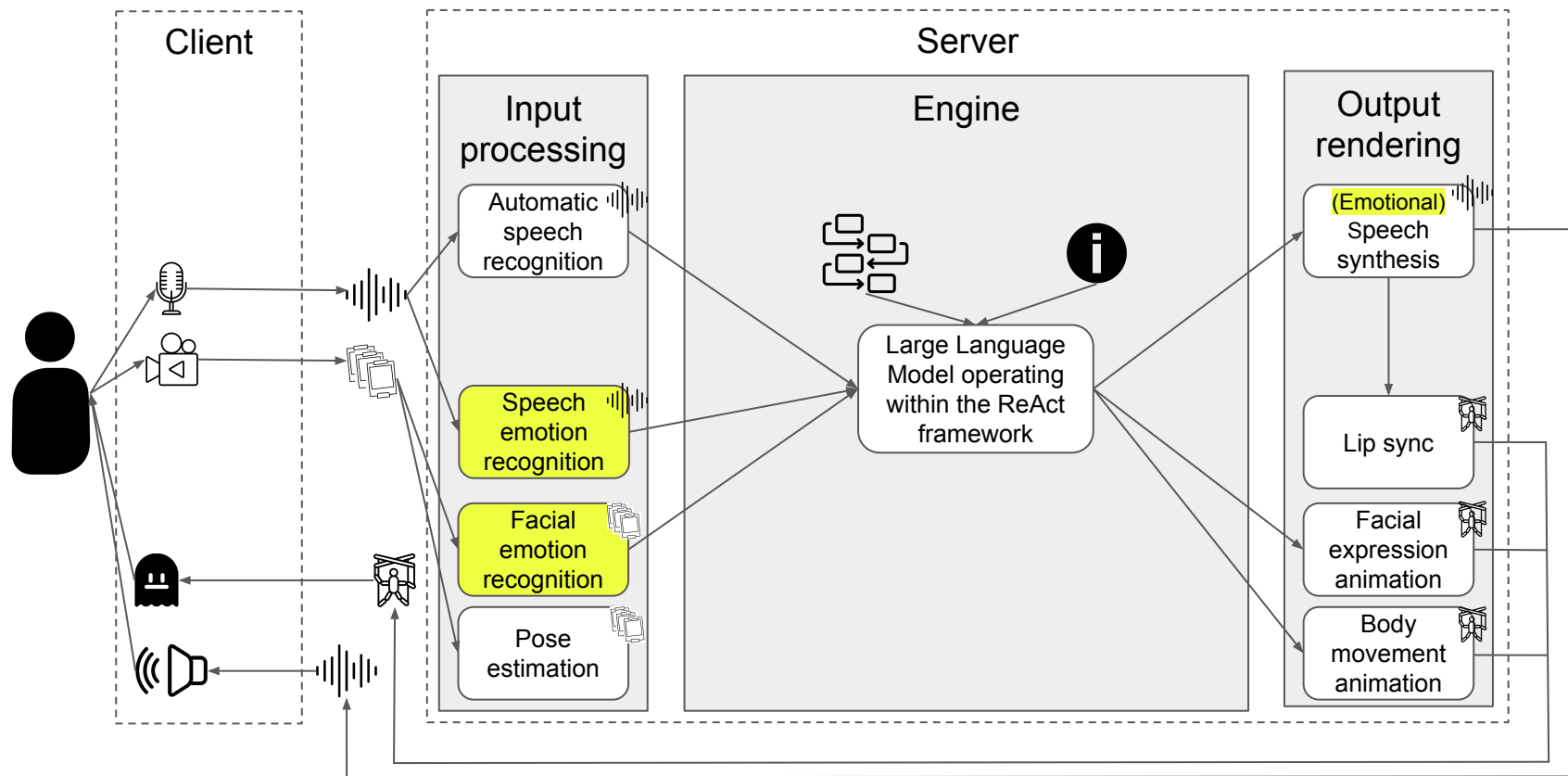
Useful resources



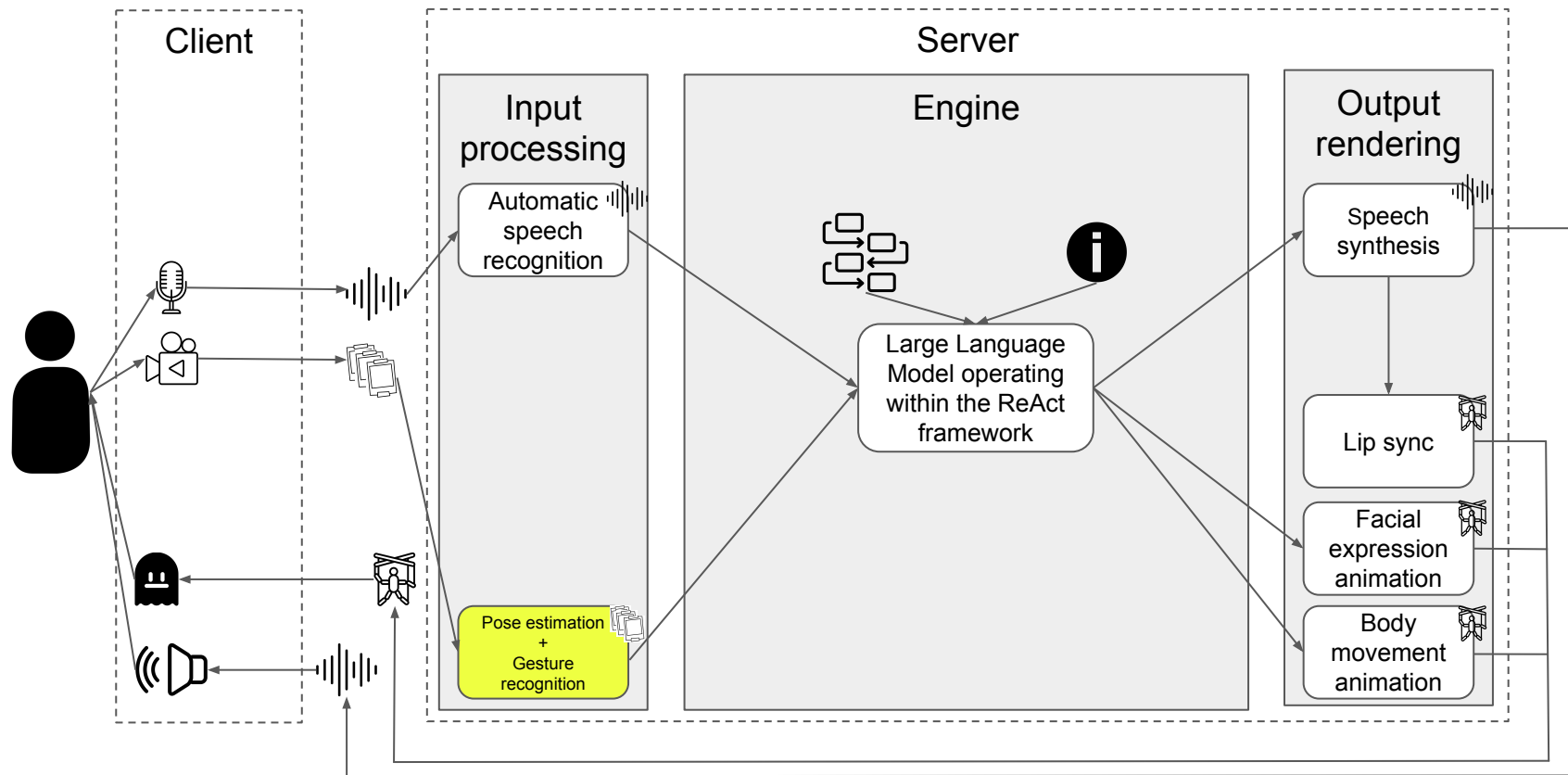
Track 2 - Ideas



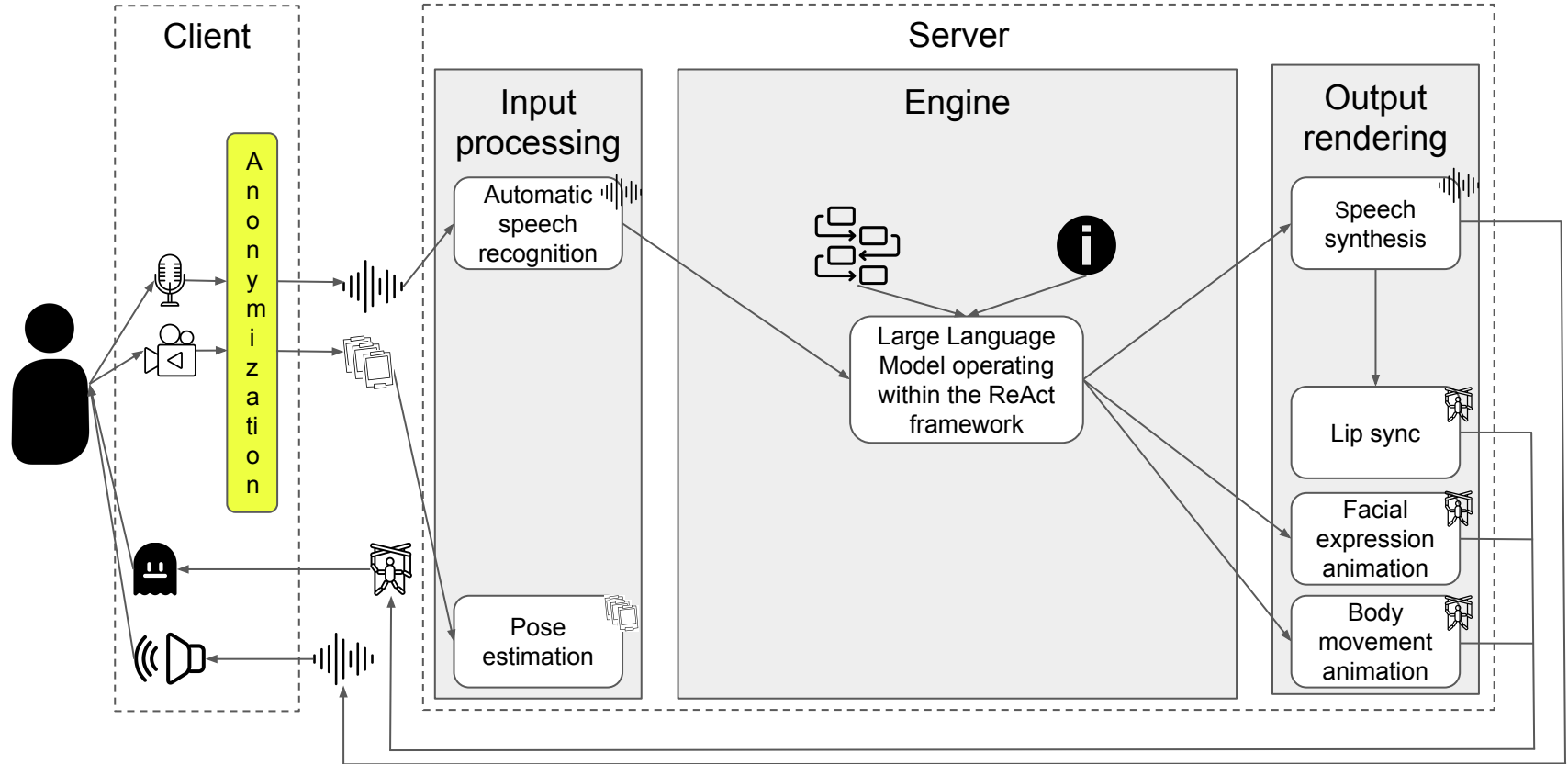
Track 2 - Ideas



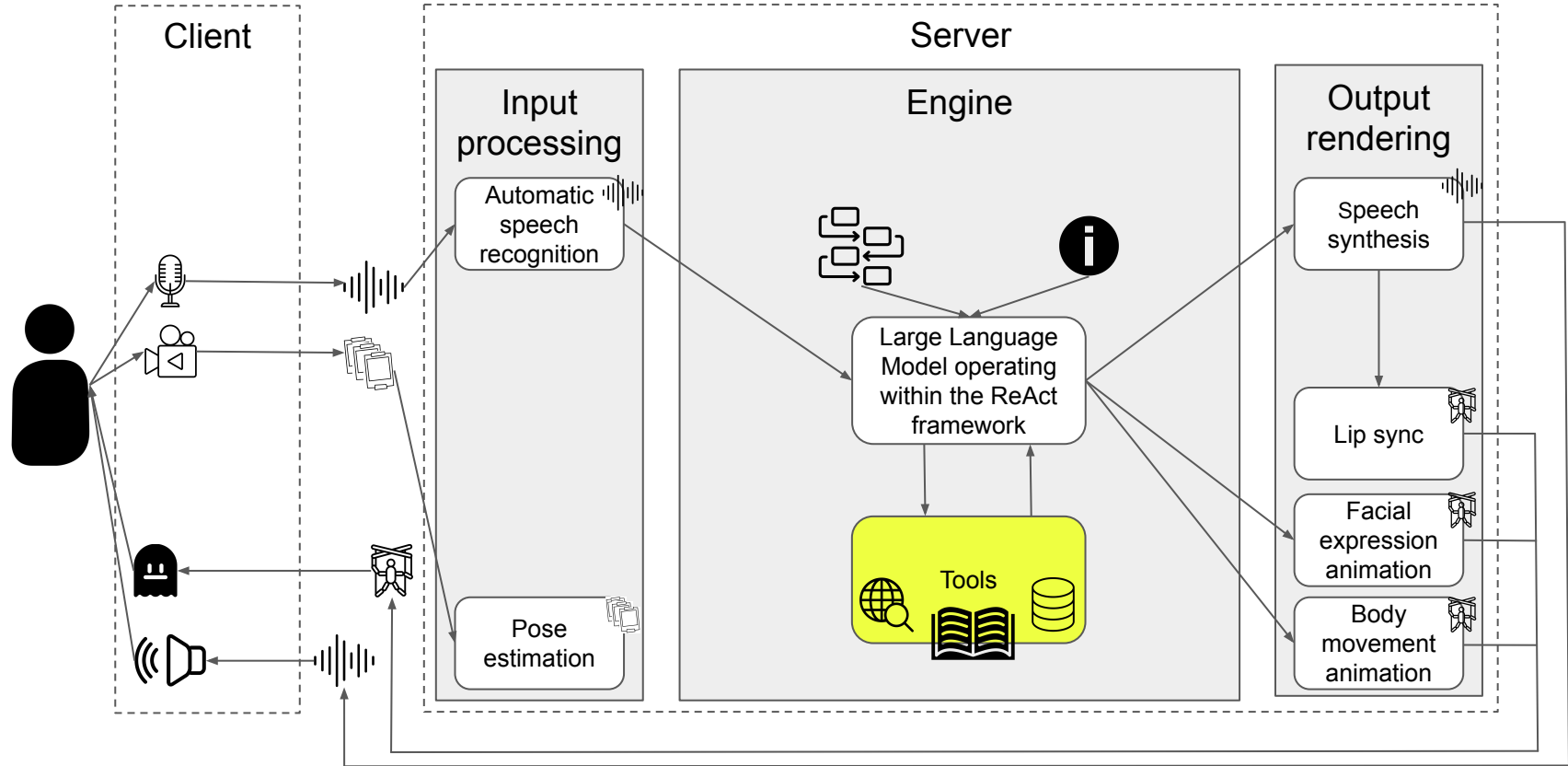
Track 2 - Ideas



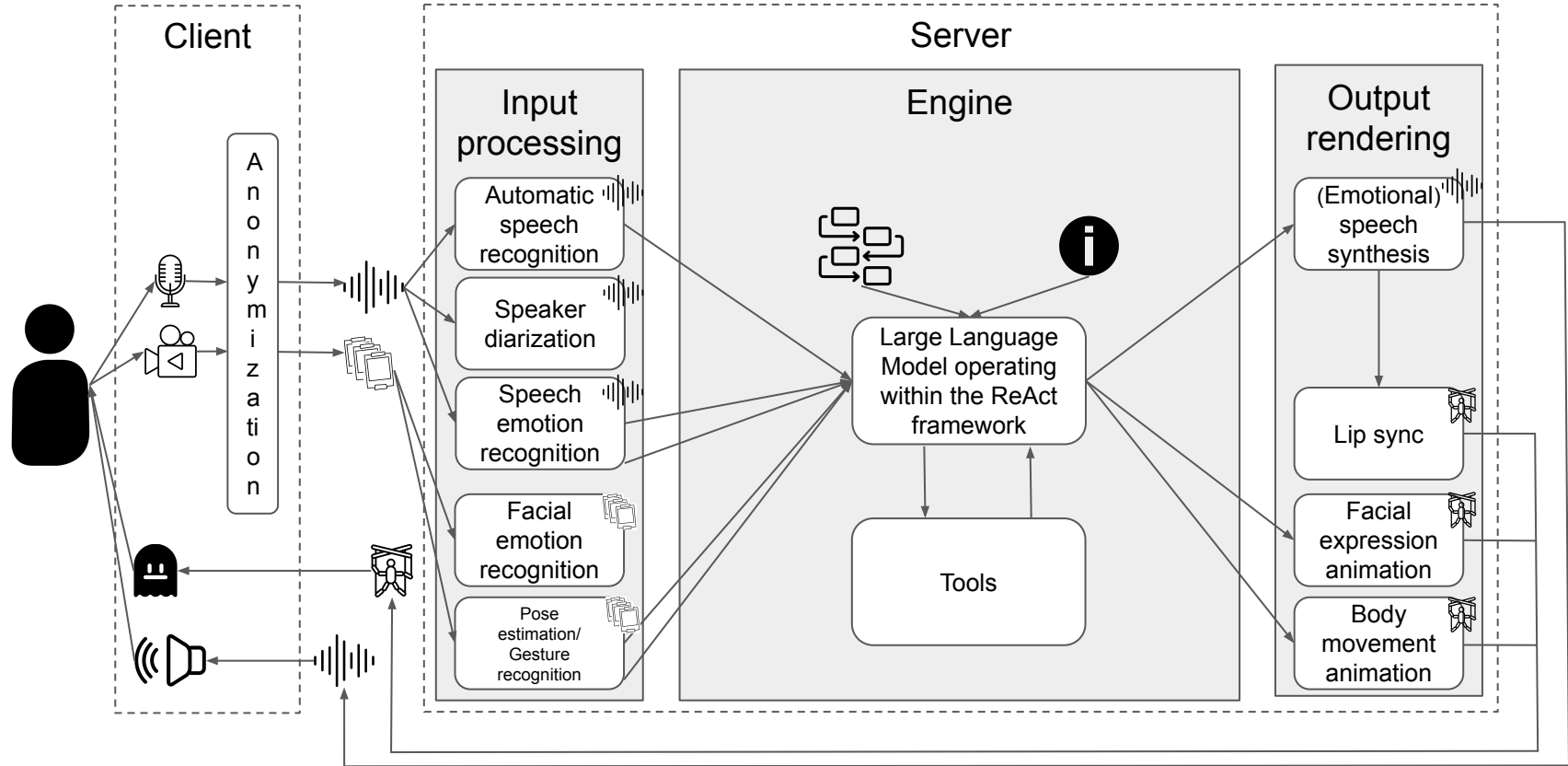
Track 2 - Ideas



Track 2 - Ideas



Track 2 - The dream



Track 2 - More ideas

 ChatGPT 5 

 Condividi 



Give me some good educational activities ideas for the AI educational hackathon - Track 2

Example Activities

- **Emotion-Aware Reading:** While a child reads aloud, the avatar senses fatigue/stress and adapts pace or encouragement.
- **Collaborative Storytelling:** Students co-create a story with the avatar, practicing expressive speech and imagination.
- **Multi-User Dialogue:** Two students interact with the avatar together, practicing turn-taking and communication.

 Fai una domanda





ChatGPT può commettere errori. Assicurati di verificare le informazioni importanti.



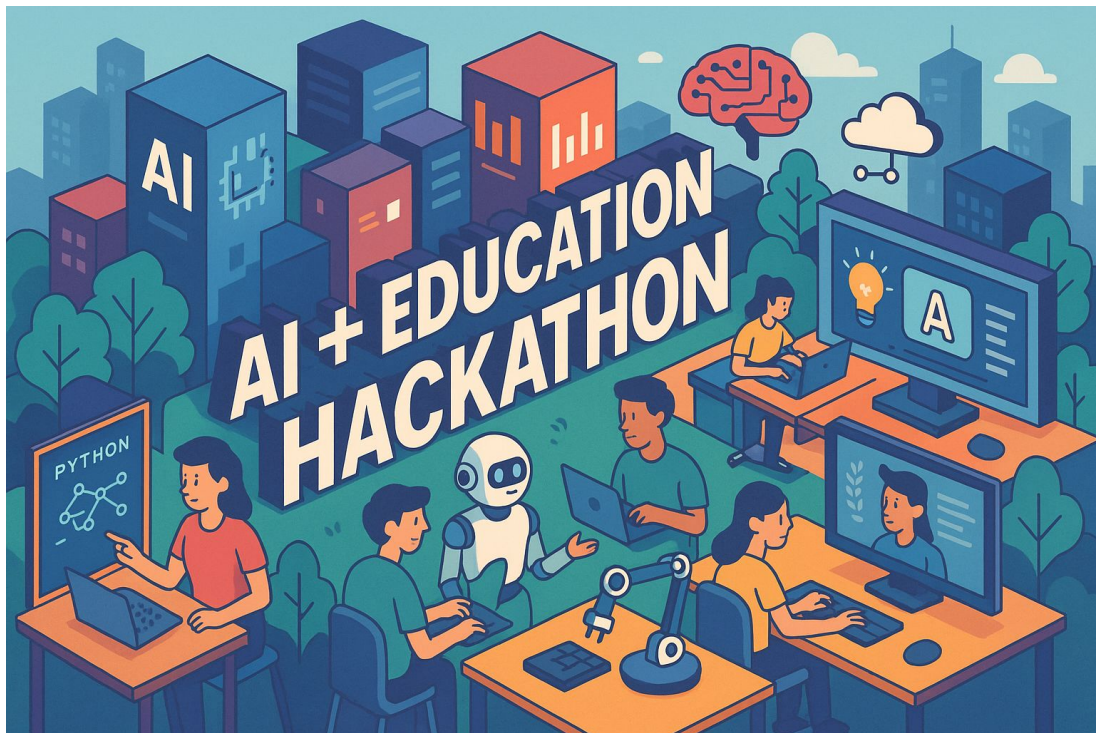
Track 3: Build your own track

Lead: Dr. Satrajit Ghosh (MIT McGovern Institute)

Goal: Define and pursue your own AI + education challenge outside the established tracks.

Focus areas:

- AI-driven assessment and content generation tools
- Accessibility innovations
- Educational platform integrations
- Cross-disciplinary projects



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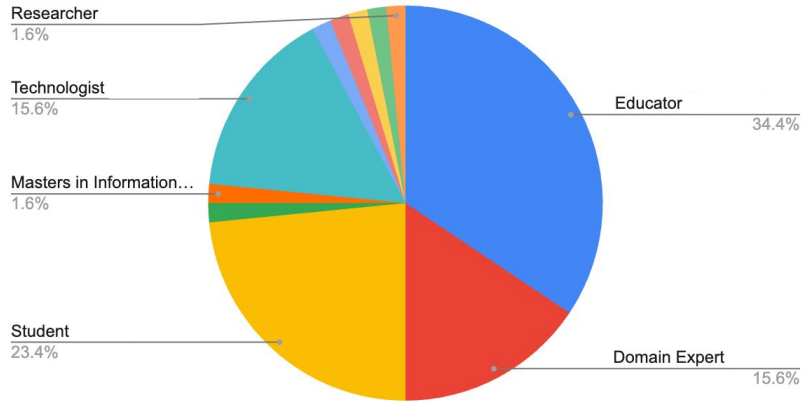
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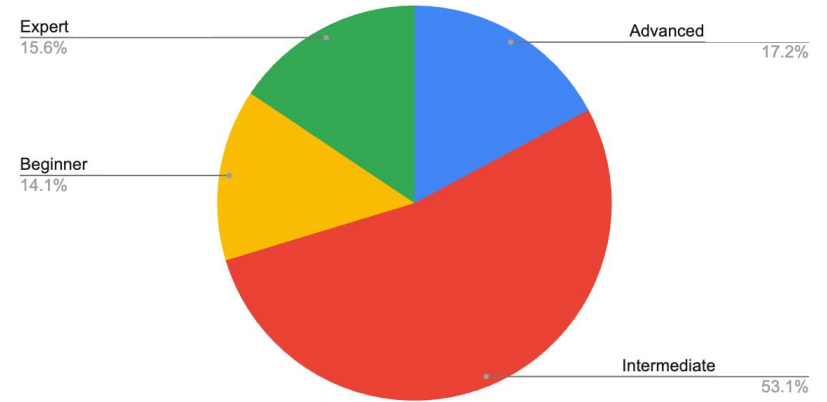
Stata Center, MIT

Participants in numbers

Count of Which category best describes your primary background?

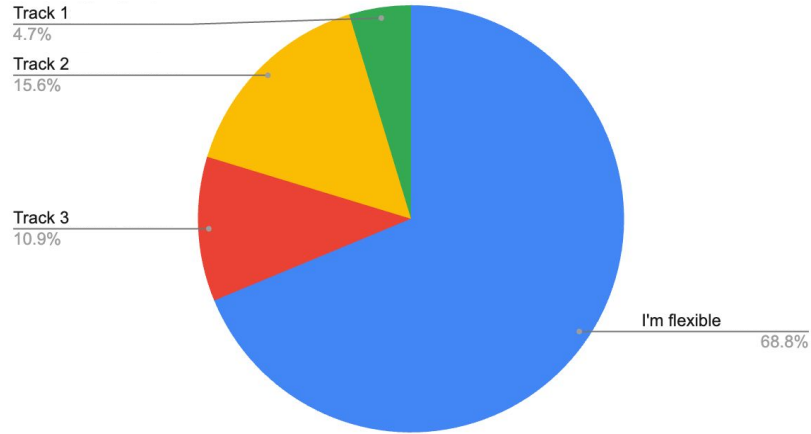


Count of What is your experience level with AI/technology in education? *

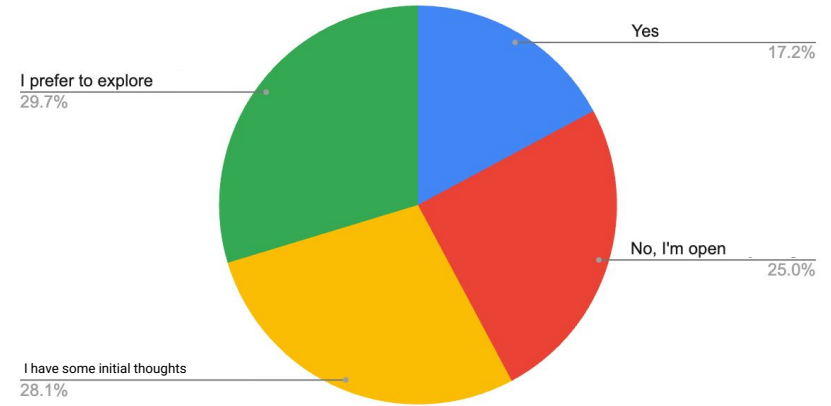


Participants in numbers

Count of Which track(s) interest you most?

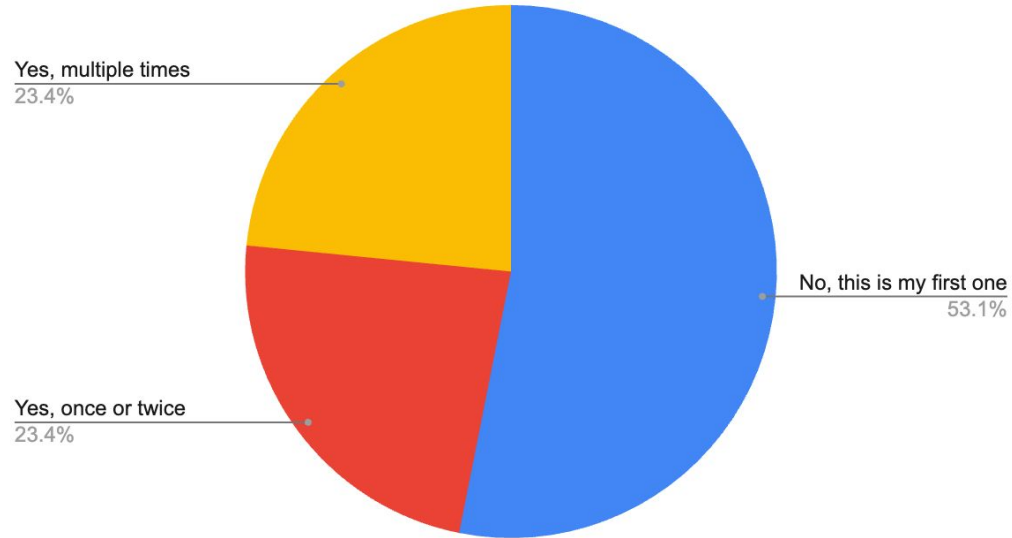


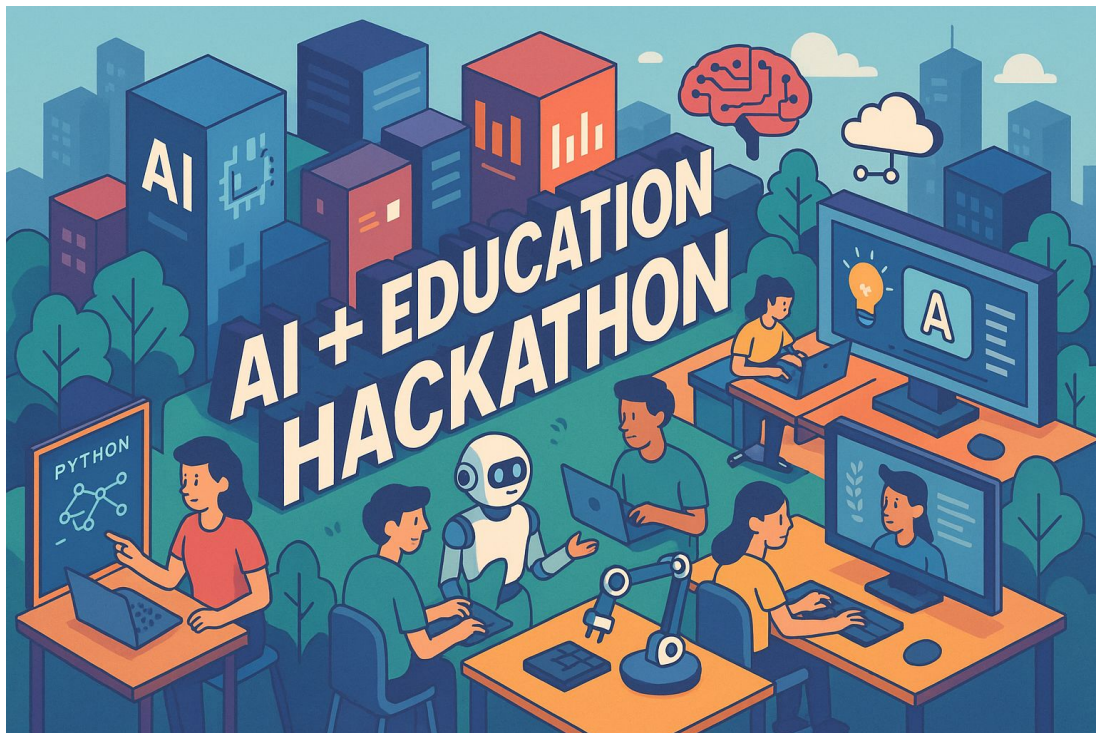
Count of Do you have a specific project idea you'd like to work on?



Participants in numbers

Count of Have you participated in a hackathon before?





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